



13-4 理想气体的等温过程和绝热过程

一 等温过程

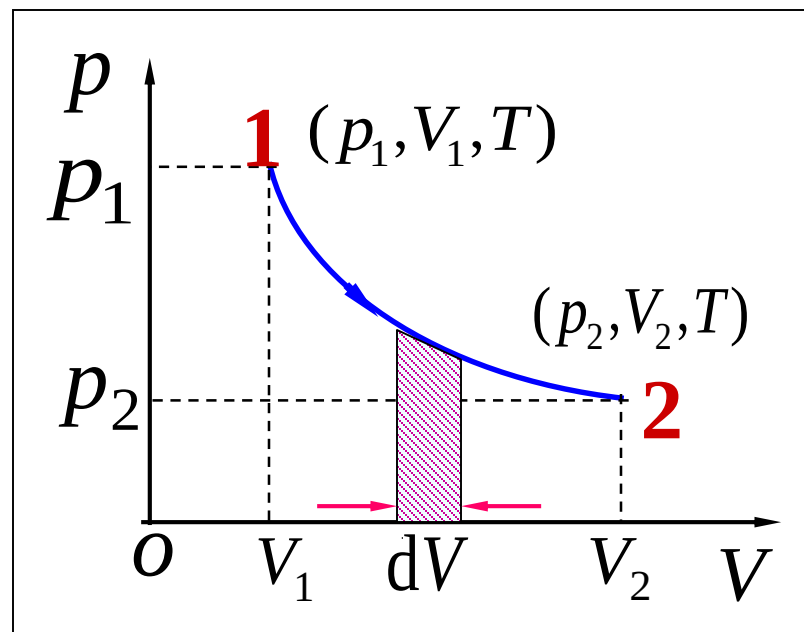
特征 $T = \text{常量}$

过程方程 $pV = \text{常量}$

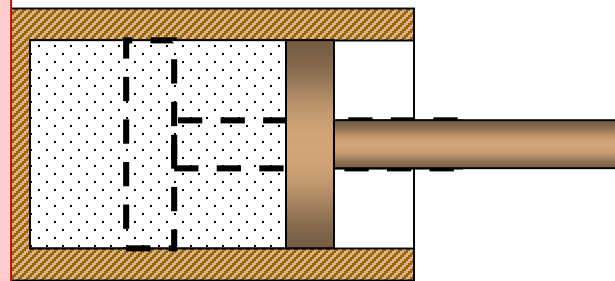
$$dE = 0$$

由热力学第一定律

$$dQ_T = dW = pdV$$



恒温热源
 T





13-4 理想气体的等温过程和绝热过程

$$Q_T = W = \int_{V_1}^{V_2} p dV$$

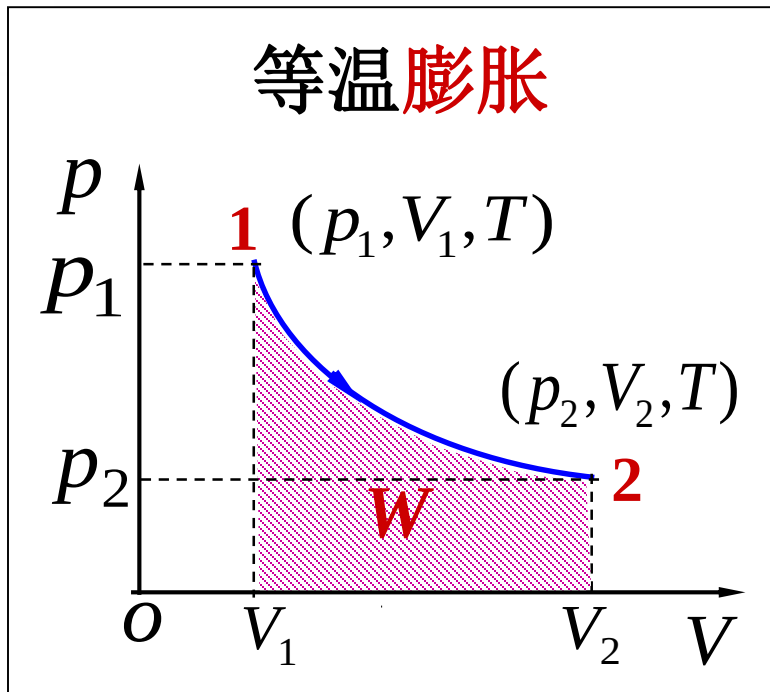
$$p = \nu \frac{RT}{V}$$

$$Q_T = W = \int_{V_1}^{V_2} \nu \frac{RT}{V} dV = \nu RT \ln \frac{V_2}{V_1}$$

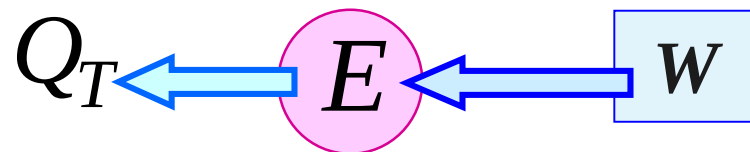
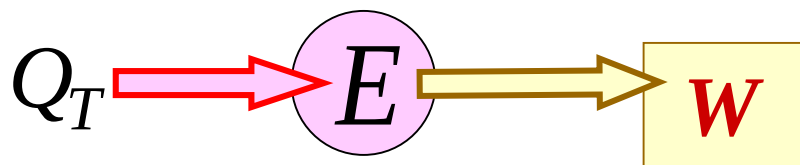
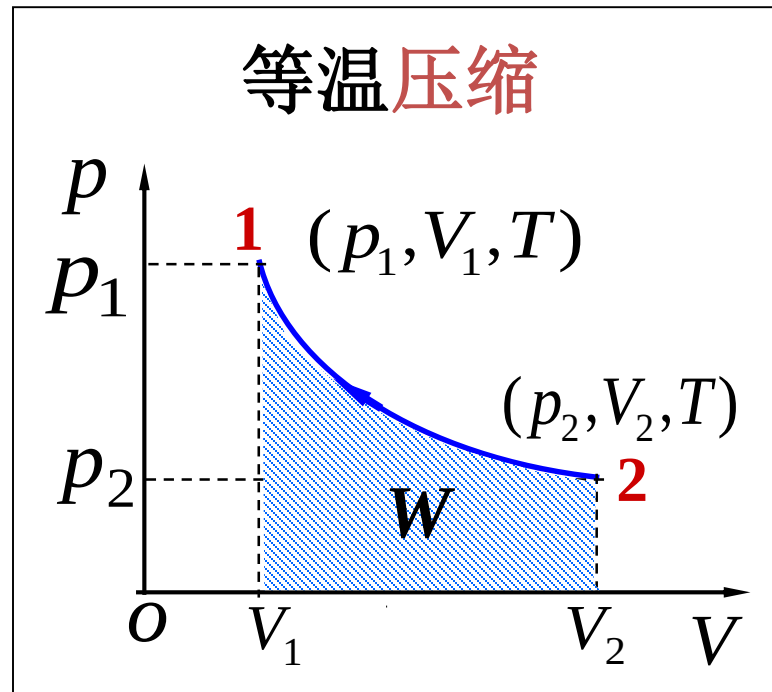


13-4 理想气体的等温过程和绝热过程

等温膨胀



等温压缩





二 绝热过程

与外界无热量交换的过程

特征 $dQ = 0$

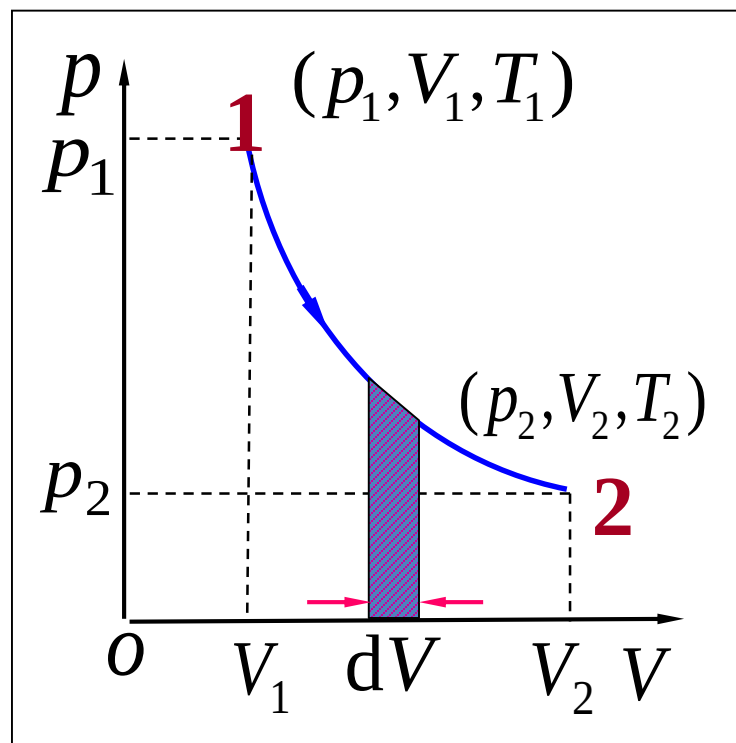
过程方程 $pV^\gamma = C$

绝热方程

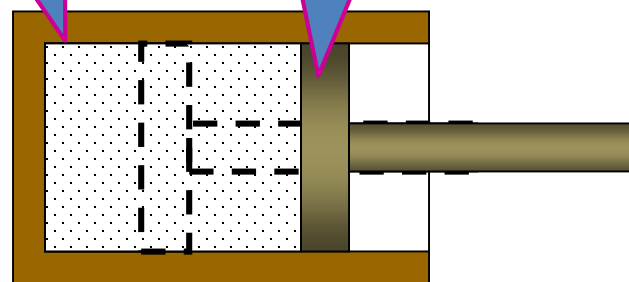
$$V^{\gamma-1}T = \text{常量}$$

$$pV^\gamma = \text{常量}$$

$$p^{\gamma-1}T^{-\gamma} = \text{常量}$$



绝热的汽缸壁和活塞





13-4 理想气体的等温过程和绝热过程

绝热过程方程的推导

$$\because dQ = 0, \quad \therefore dE + dW = 0$$

$$\nu C_{V,m} dT + p dV = 0$$

$$pV = \nu RT$$

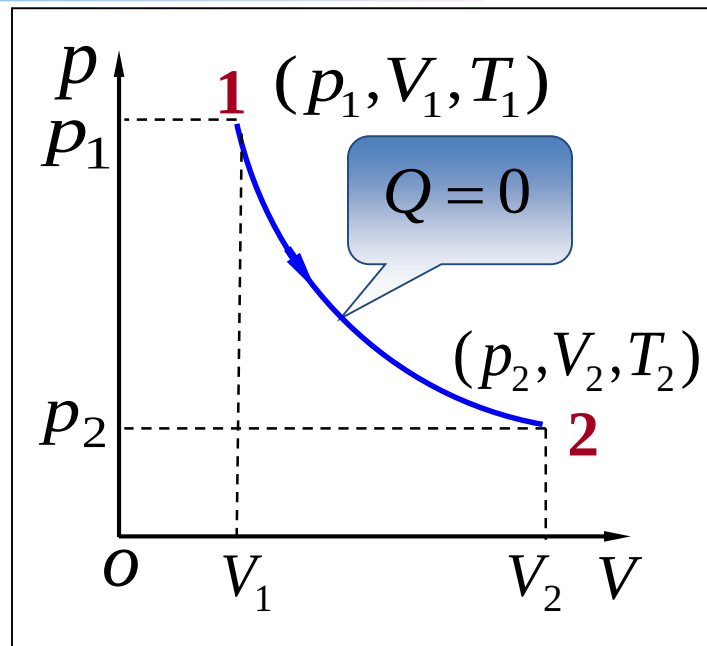
$$pdV + Vdp = \nu R dT$$

$$C_{V,m} (pdV + Vdp) + R \cdot pdV = 0$$

$$C_{p,m} pdV + C_{V,m} Vdp = 0 \quad \gamma pdV + Vdp = 0$$

$$\gamma \frac{dV}{V} + \frac{dp}{p} = d(\ln pV^\gamma) = 0$$

$$pV^\gamma = C$$





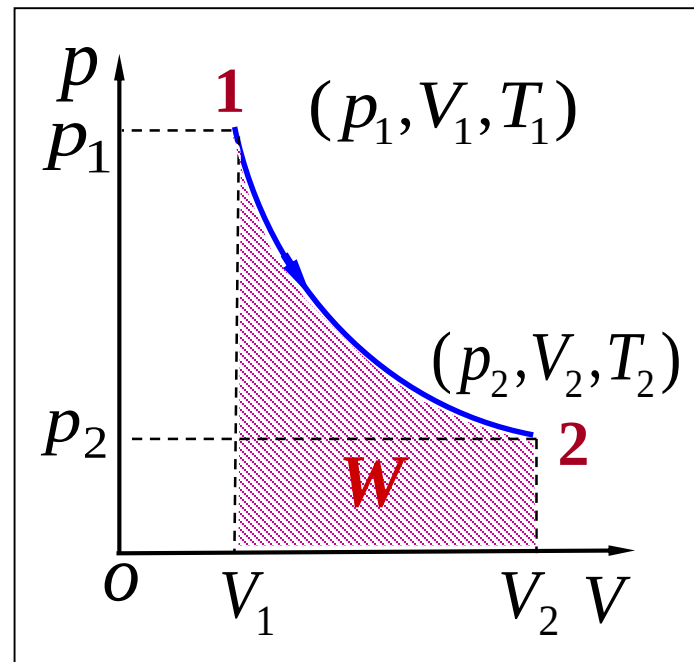
绝热过程的功

由热力学第一定律有 $Q = \Delta E + W = 0$

$$W = -\Delta E = \nu C_{V,m} (T_1 - T_2)$$

由 $pV = \nu RT$ 可得

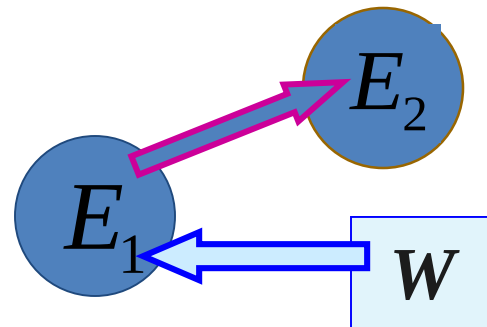
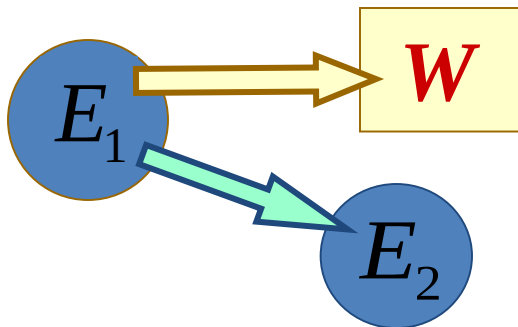
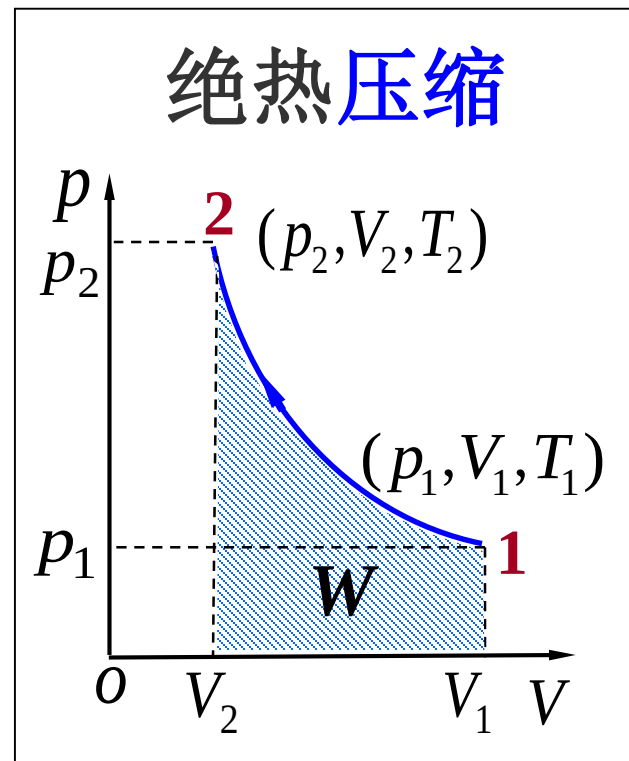
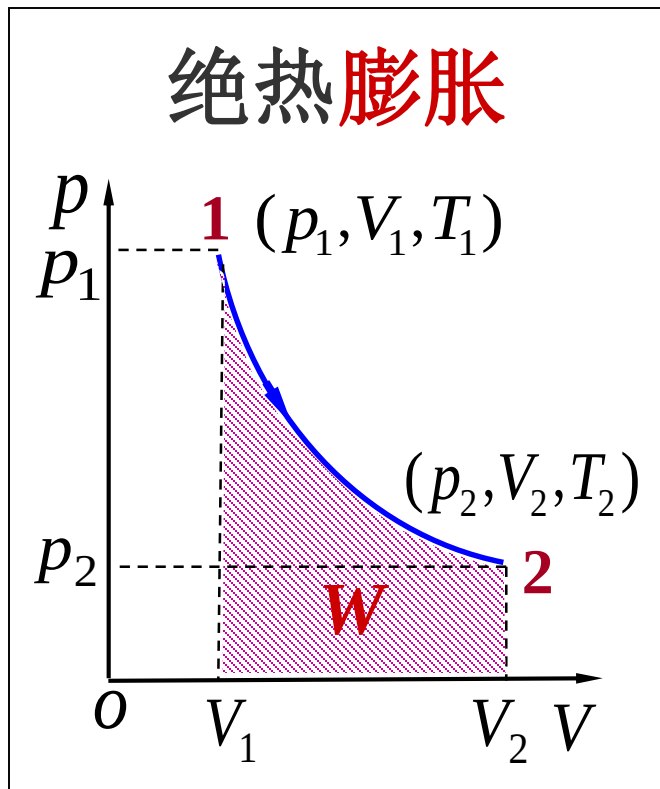
$$\begin{aligned} W &= C_{V,m} \left(\frac{p_1 V_1}{R} - \frac{p_2 V_2}{R} \right) \\ &= \frac{C_{V,m}}{C_{p,m} - C_{V,m}} (p_1 V_1 - p_2 V_2) \end{aligned}$$



$$W = -\frac{p_2 V_2 - p_1 V_1}{\gamma - 1}$$



13-4 理想气体的等温过程和绝热过程





三 绝热线和等温线

绝热过程曲线的斜率

$$pV^\gamma = \text{常量}$$

$$\gamma p V^{\gamma-1} dV + V^\gamma dp = 0$$

$$\left(\frac{dp}{dV}\right)_a = -\gamma \frac{p_A}{V_A}$$



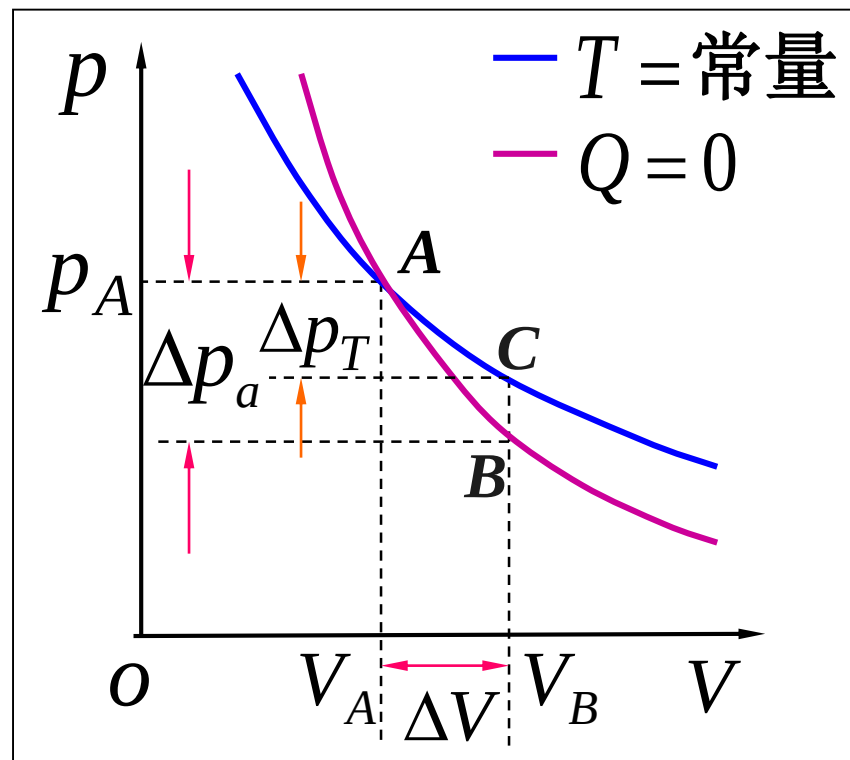
13-4 理想气体的等温过程和绝热过程

等温过程曲线的斜率

$$pV = \text{常量}$$

$$pdV + Vdp = 0$$

$$\left(\frac{dp}{dV}\right)_T = -\frac{p_A}{V_A}$$

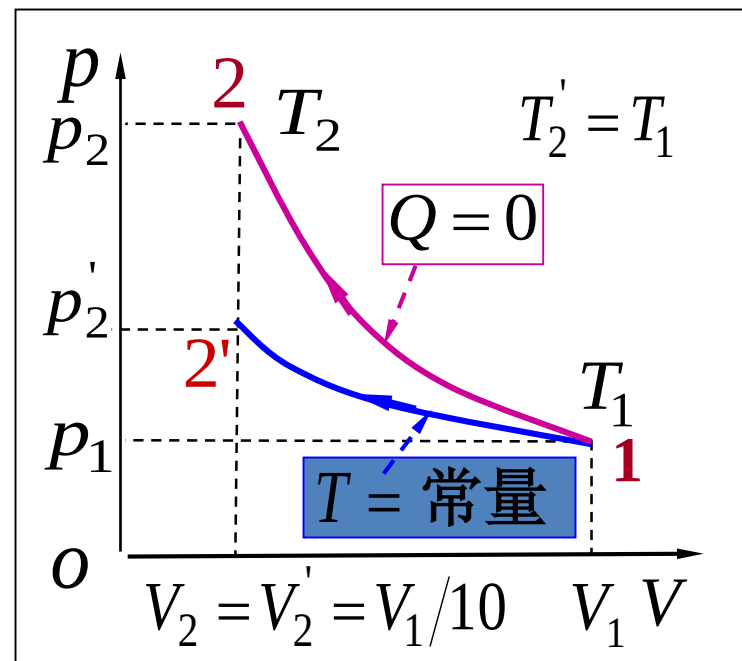


绝热线的斜率大于等温线的斜率.



13-4 理想气体的等温过程和绝热过程

例1 设有 5 mol 的氢气，最初温度 20°C ，压强 $1.013 \times 10^5 \text{ Pa}$ ，求下列过程中把氢气压缩为原体积的 $1/10$ 需作的功：(1) 等温过程 (2) 绝热过程 (3) 经这两过程后，气体的压强各为多少？





13-4 理想气体的等温过程和绝热过程

已知: $\nu = 5 \text{ mol}$ $T_0 = 293 \text{ K}$

$$P_0 = 1.013 \times 10^5 \text{ Pa} \quad V = 0.1 V_0$$

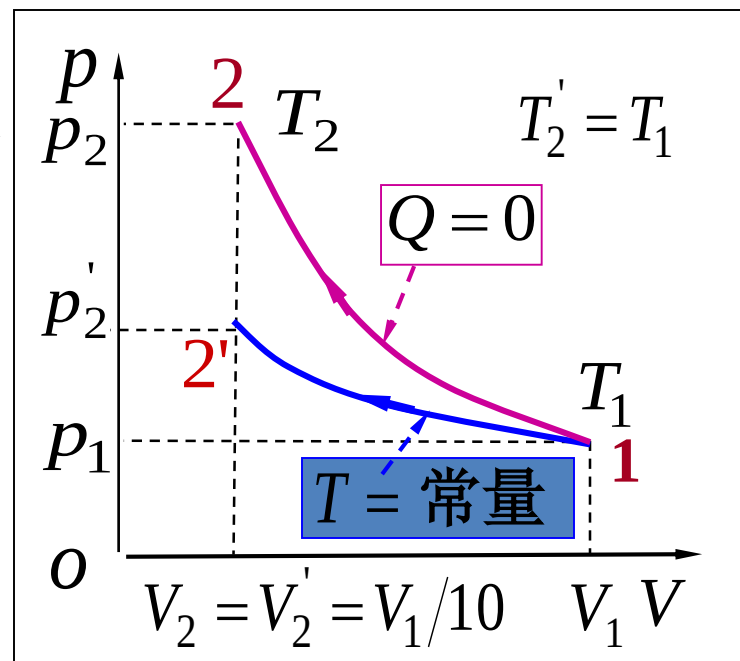
解 (1) 等温过程

$$W'_{12} = \nu RT \ln \frac{V'_2}{V_1} = -2.80 \times 10^4 \text{ J}$$

(2) 氢气为双原子气体

由表查得 $\gamma = 1.41$, 有

$$T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1} = 753 \text{ K}$$





13-4 理想气体的等温过程和绝热过程

$$W_{12} = -\nu C_{V,m}(T_2 - T_1) \quad C_{V,m} = 20.44 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$$

$$W_{12} = -4.70 \times 10^4 \text{ J}$$

(3) 对等温过程

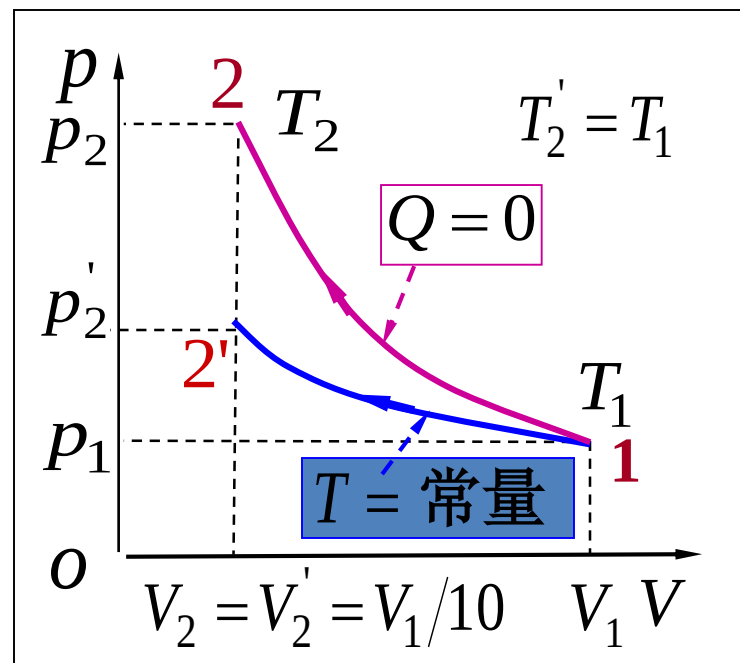
$$p'_2 = p_1 \left(\frac{V_1}{V_2} \right)$$

$$= 1.01 \times 10^6 \text{ Pa}$$

对绝热过程, 有

$$p_2 = p_1 \left(\frac{V_1}{V_2} \right)^\gamma$$

$$= 2.55 \times 10^6 \text{ Pa}$$





四、多方过程(polytropic process)

$$C_{n,m} = \text{Const.}$$

多方过程方程: $pV^n = C$

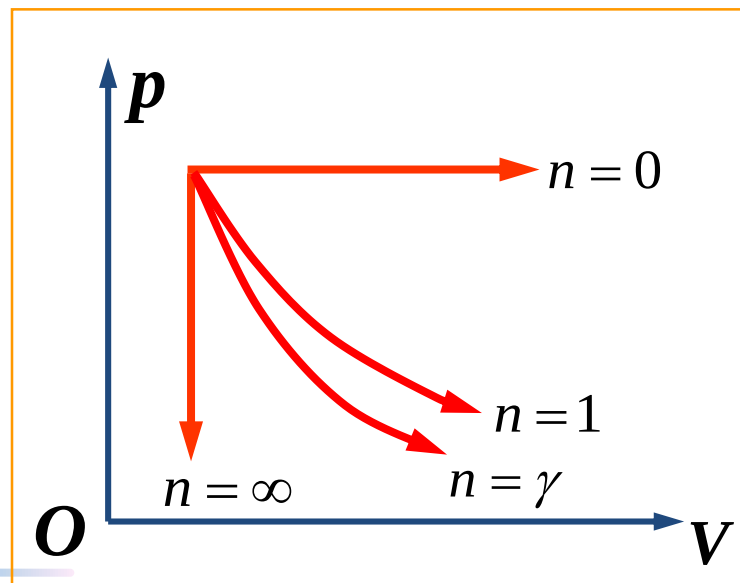
其中 $n = \frac{C_{n,m} - C_{p,m}}{C_{n,m} - C_{V,m}}$ 为多方指数

等压过程: $dp = 0$ $n = 0$

等体过程: $dV = 0$ $n \rightarrow \infty$

等温过程: $dT = 0$ $n = 1$

绝热过程: $Q = 0$ $n = \gamma$





13-4 理想气体的等温过程和绝热过程

- 计算多方过程的功。

$$pV^n = C$$

$$p = CV^{-n}$$

$$W = \int p dV = \int CV^{-n} dV = \frac{1}{-n+1} \int dV^{-n+1}$$

$$W = \frac{C}{1-n} (V_f^{1-n} - V_i^{1-n})$$

$$C = p_f V_f^n = p_i V_i^n$$

$$W = -\frac{1}{n-1} (p_f V_f - p_i V_i)$$

$$pV = \nu RT$$

$$W = -\frac{\nu R}{n-1} \Delta T$$

$$dW = \nu \frac{R}{1-n} dT$$



13-4 理想气体的等温过程和绝热过程

• 多方过程的热容量 $dQ_n = \nu C_{n,m} dT$

$$dE = \nu \frac{i}{2} R dT = \nu C_{V,m} dT \quad dW = \nu \frac{R}{1-n} dT$$

$$dQ = dE + dW$$

$$C_{n,m} = C_{V,m} + \frac{R}{1-n} \quad C_{p,m} = C_{V,m} + R$$

$$C_{n,m} = C_{V,m} + \frac{C_{p,m} - C_{V,m}}{1-n} \quad \gamma = C_{p,m} / C_{V,m}$$

$$C_{n,m} = C_{V,m} \left(1 + \frac{\gamma - 1}{1 - n} \right) = \frac{\gamma - n}{1 - n} C_{V,m}$$



一、热力学循环过程

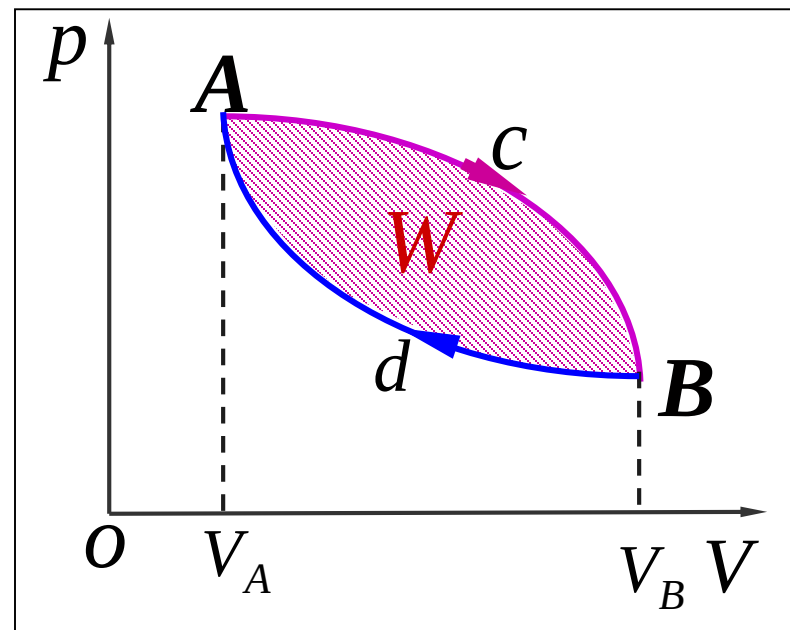
1、循环过程

系统经过一系列变化状态过程后，又回到原来的状态的过程叫热力学循环过程。

特征 $\Delta E = 0$

由热力学第一定律

$$Q = W$$





净功 $W = Q_1 - Q_2 = Q$

总吸热 $\longrightarrow Q_1$

总放热 $\longrightarrow Q_2$ (取绝对值)

净吸热 $\longrightarrow Q$

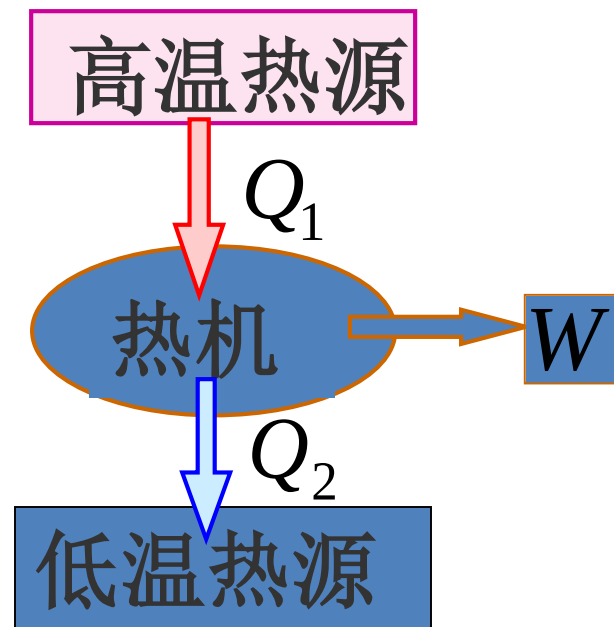
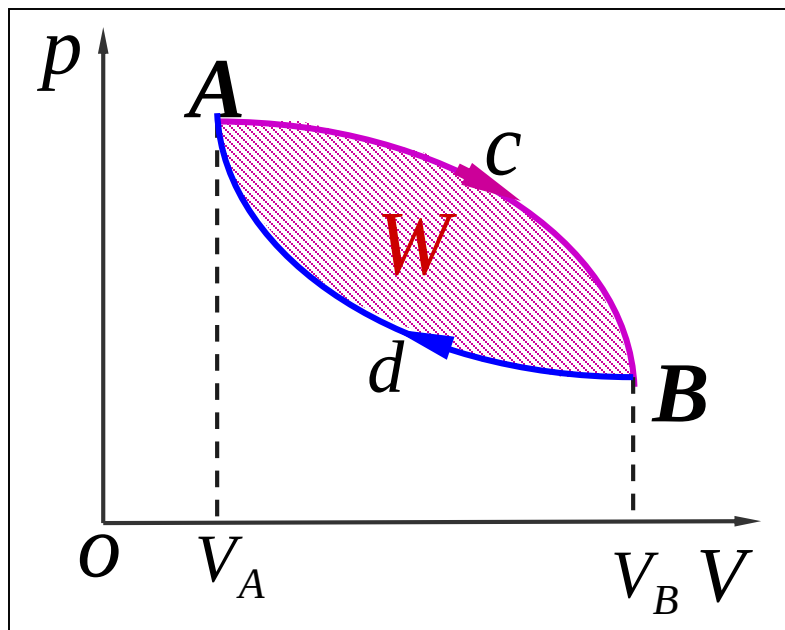
2 热机效率和致冷机的致冷系数

热机 (正循环) $W > 0$

致冷机 (逆循环) $W < 0$



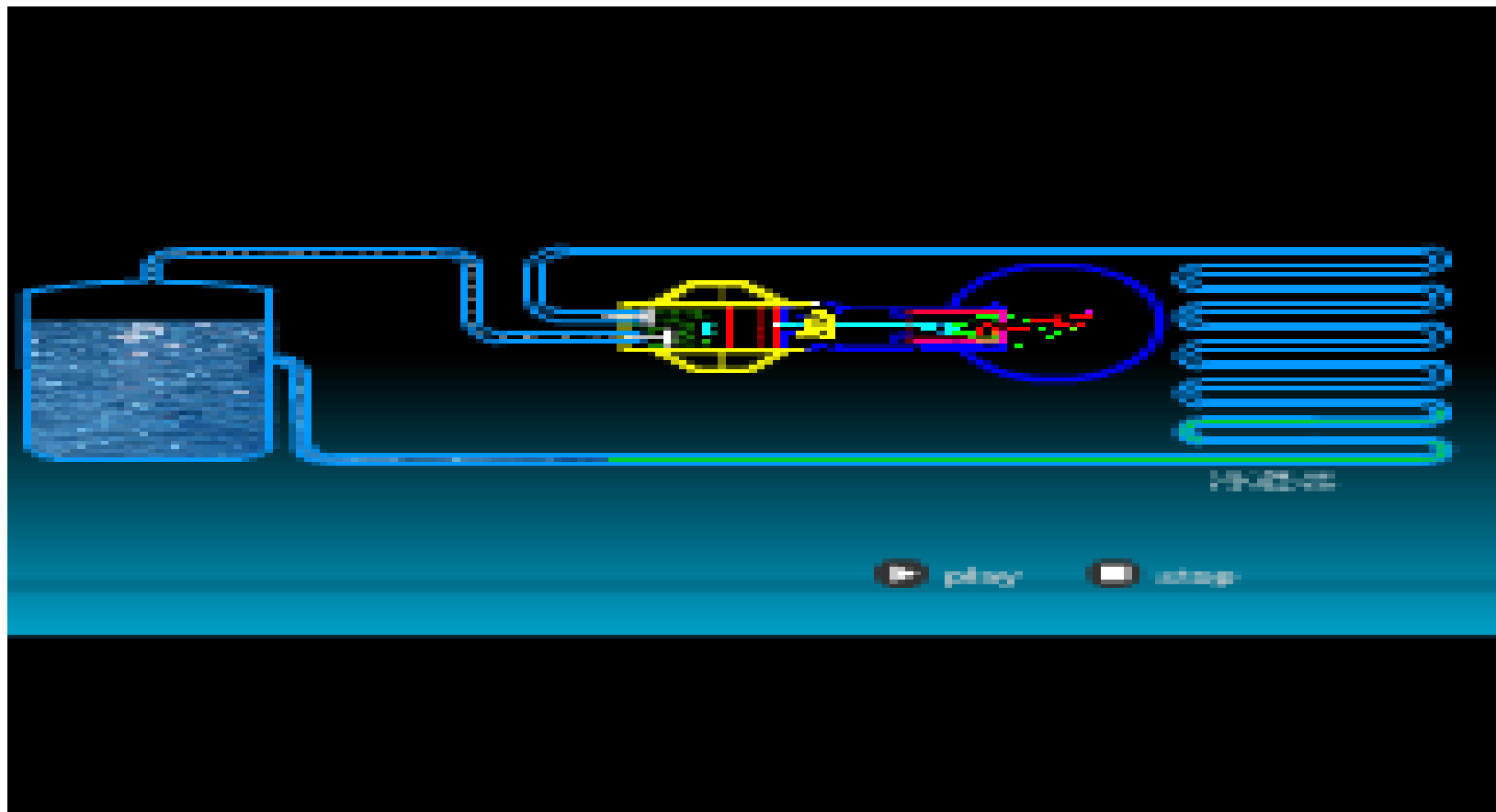
13-5 循环过程 卡诺循环



热机效率 $\eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$



热机：持续地将热量转变为功的机器。





各种热机的效率

液体燃料火箭 $\eta = 48\%$

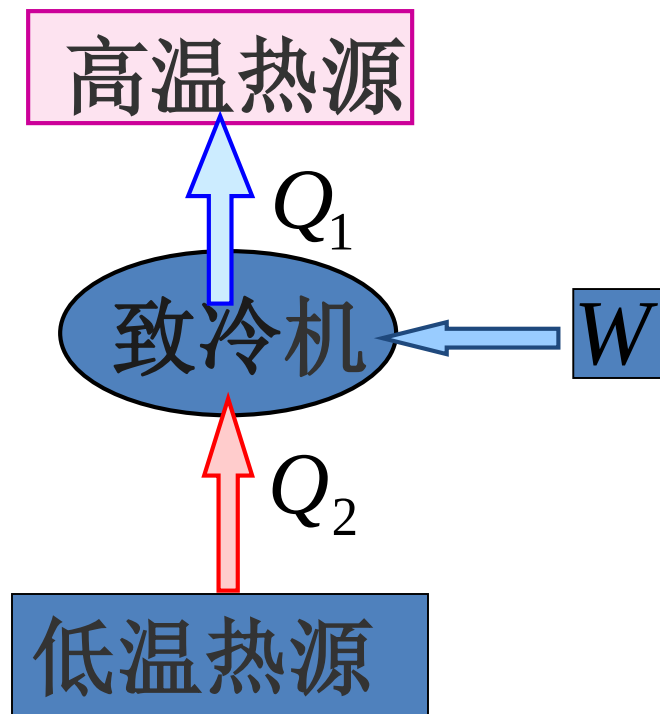
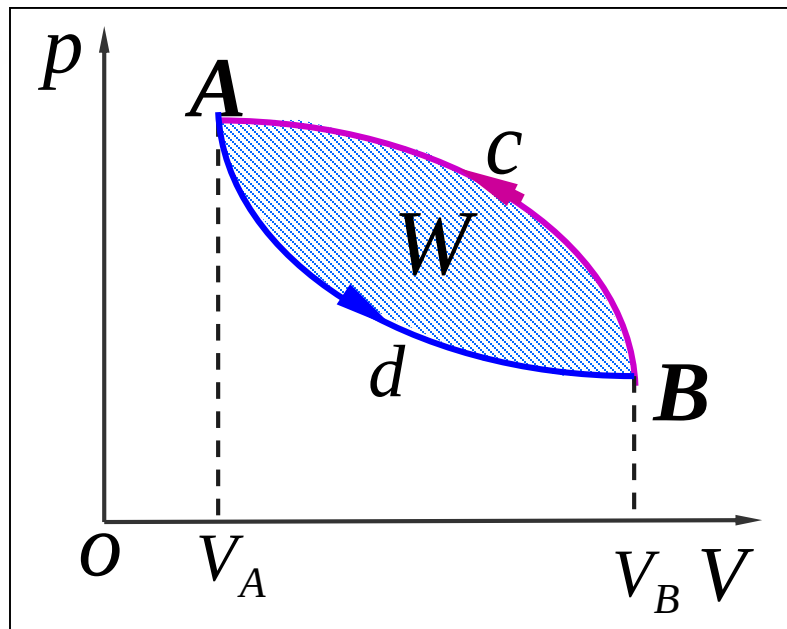
柴油机 $\eta = 37\%$

汽油机 $\eta = 25\%$

蒸汽机 $\eta = 8\%$



13-5 循环过程 卡诺循环

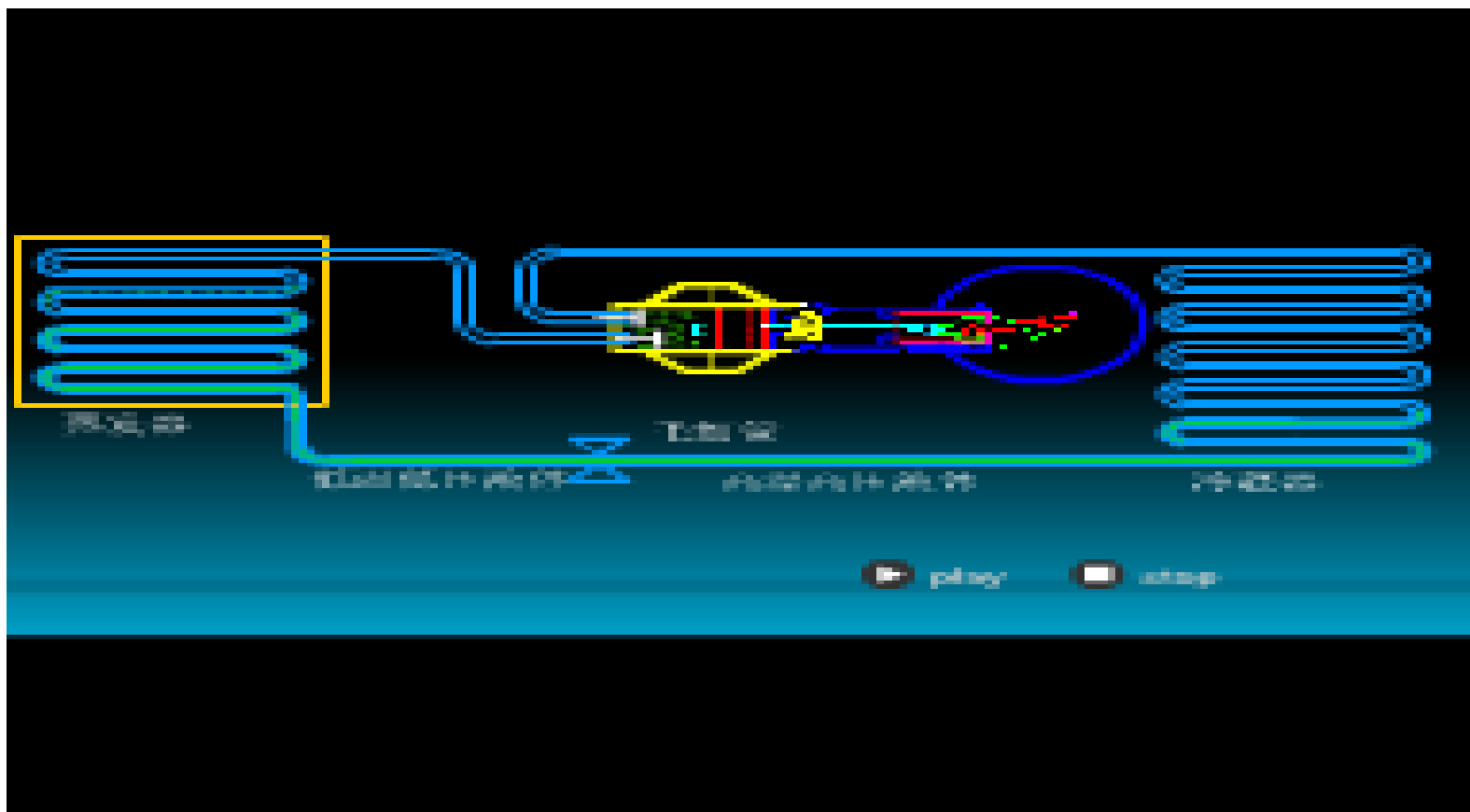


制冷机致冷系数

$$e = \frac{Q_2}{|W|} = \frac{Q_2}{Q_1 - Q_2}$$



冰箱循环示意图





三 卡诺循环

1698年萨维利和1705年纽可门先后发明了蒸汽机，当时蒸汽机的效率极低。1765年瓦特进行了重大改进，大大提高了效率。

1824年法国的年青工程师卡诺提出一个工作在两热源之间的理想循环——卡诺循环。给出了热机效率的理论极限值；他还提出了著名的卡诺定理。

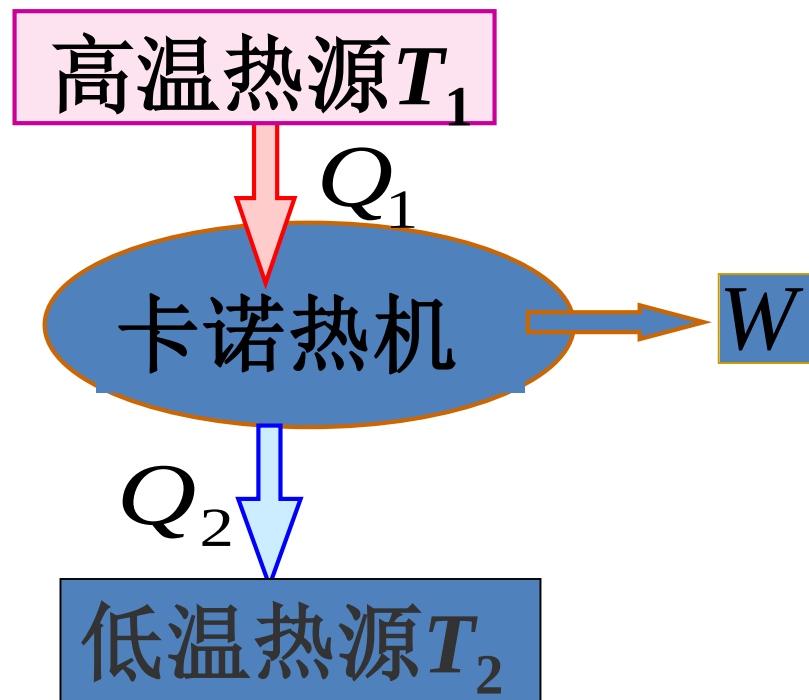
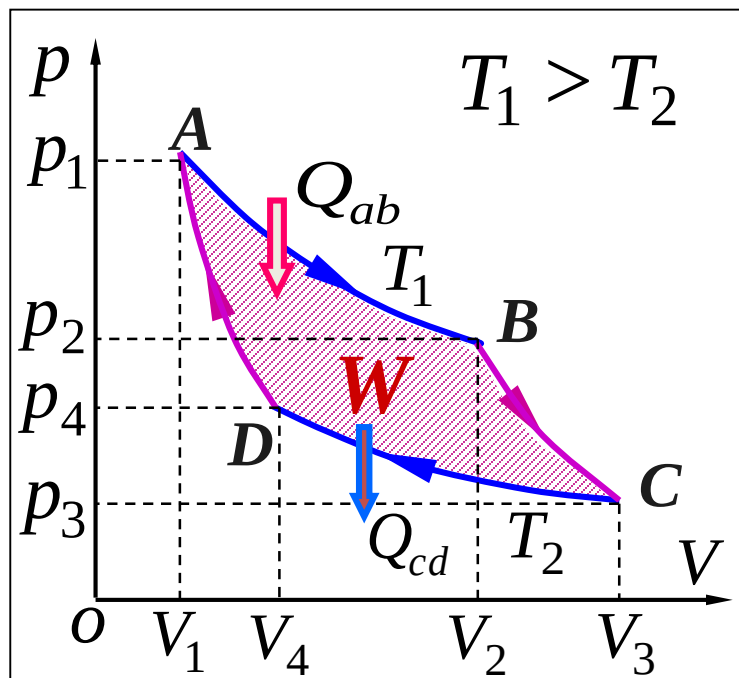
卡诺循环由两个等温过程和两个绝热过程组成。

$A \rightarrow B$ 等温膨胀

$B \rightarrow C$ 绝热膨胀

$C \rightarrow D$ 等温压缩

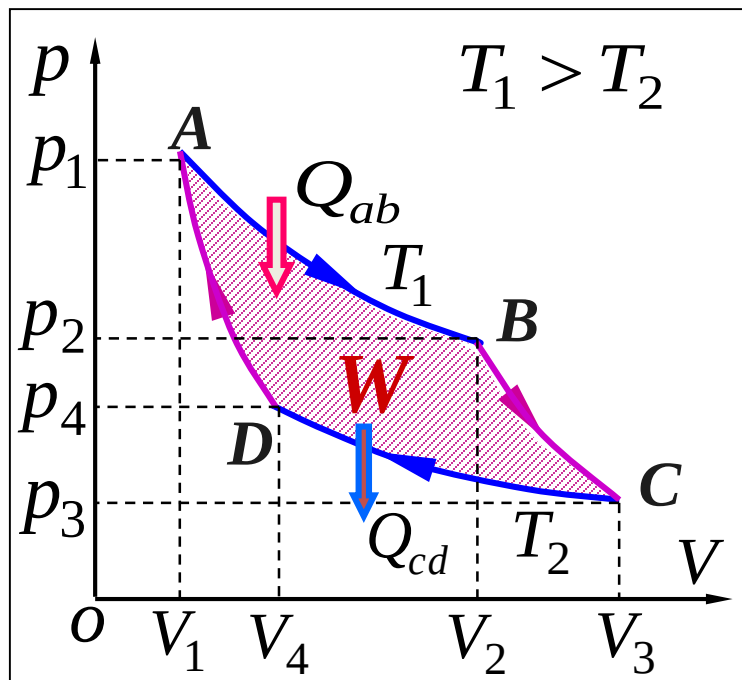
$D \rightarrow A$ 绝热压缩





A — B 等温膨胀吸热

C — D 等温压缩放热



$$\eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}$$

13-5 循环过程 卡诺循环

$$Q_1 = Q_{ab} = \nu R T_1 \ln \frac{V_2}{V_1}$$
$$Q_2 = |Q_{cd}| = \nu R T_2 \ln \frac{V_3}{V_4}$$

B — C 绝热过程

$$T_1 V_2^{\gamma-1} = T_2 V_3^{\gamma-1}$$

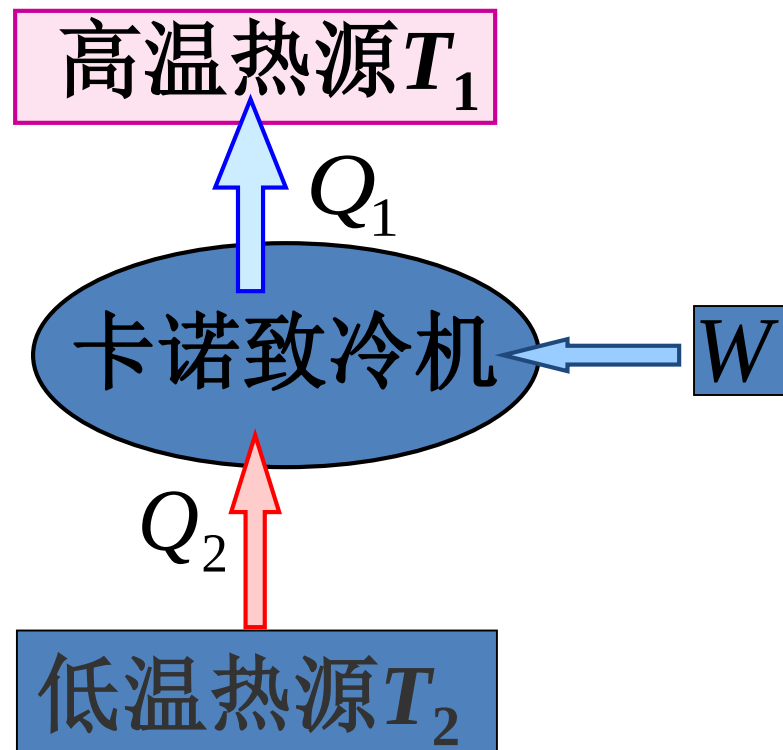
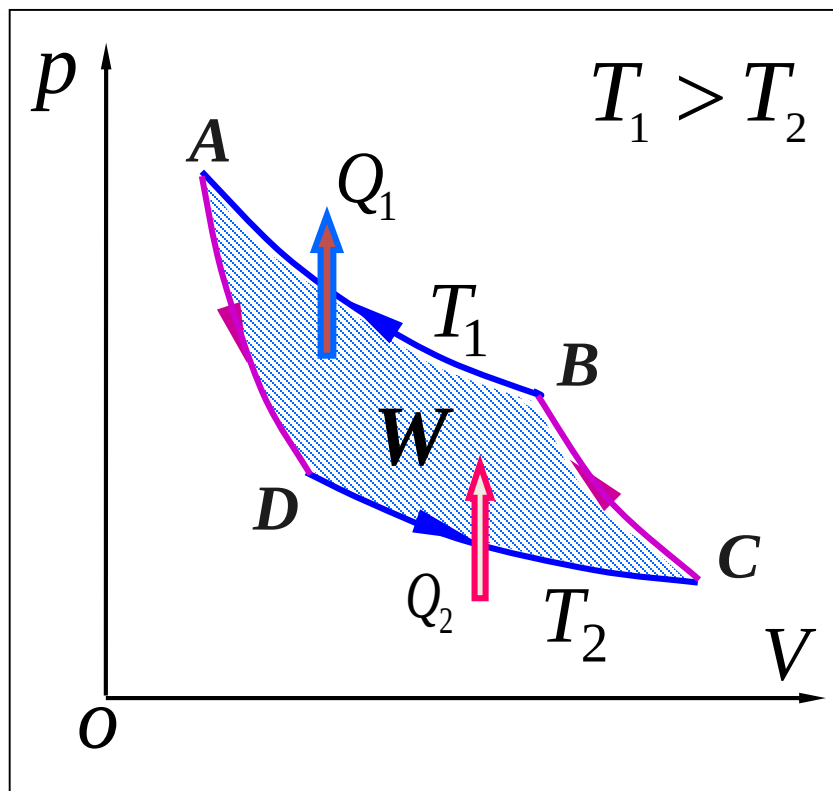
D — A 绝热过程

$$T_1 V_1^{\gamma-1} = T_2 V_4^{\gamma-1}$$

$$\frac{V_2}{V_1} = \frac{V_3}{V_4}$$



◆ 卡诺致冷机（卡诺逆循环）



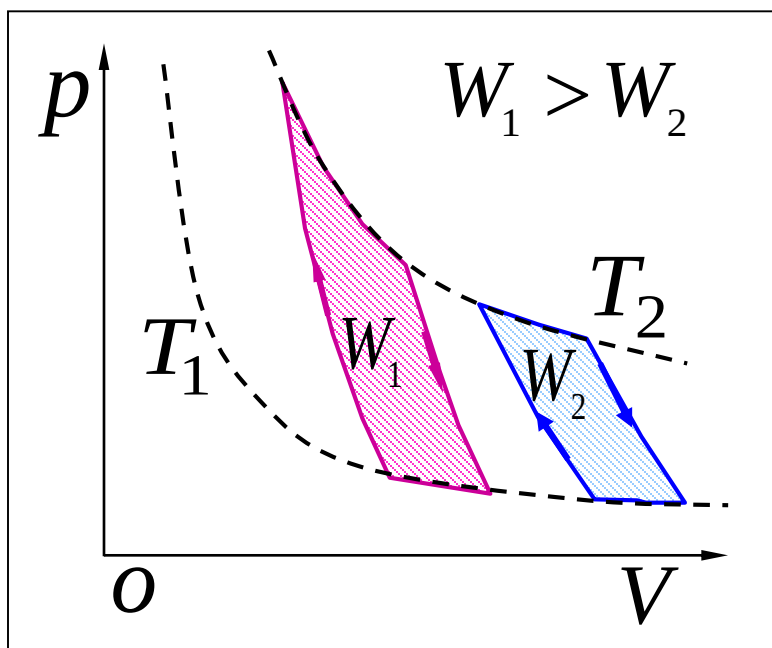
卡诺致冷机致冷系数

$$e = \frac{Q_2}{Q_1 - Q_2} = \frac{T_2}{T_1 - T_2}$$

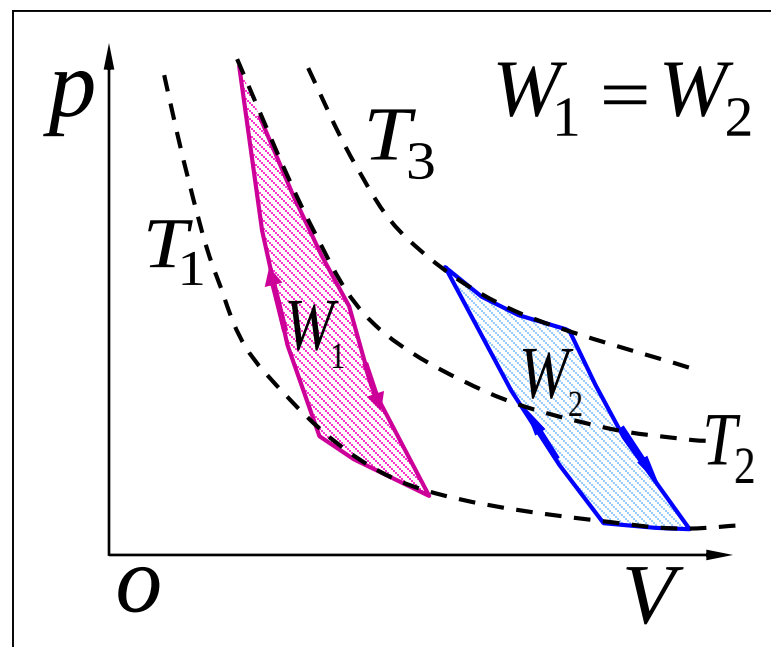


讨 论

图中两卡诺循环 $\eta_1 = \eta_2$ 吗？



$$\eta_1 = \eta_2$$



$$\eta_1 < \eta_2$$



例2 一电冰箱放在室温为 20°C 的房间里，冰箱储藏柜中的温度维持在 5°C 。现每天有 $2.0 \times 10^7 \text{ J}$ 的热量自房间传入冰箱内，若要维持冰箱内温度不变，外界每天需作多少功，其功率为多少？设在 5°C 至 20°C 之间运转的冰箱的致冷系数是卡诺致冷机致冷系数的 55%。



解： $e = e_{\text{卡}} \times 55\% = \frac{T_2}{T_1 - T_2} \times \frac{55}{100} = 10.2$

房间传入冰箱的热量 $Q' = 2.0 \times 10^7 \text{ J}$

热平衡时 $Q' = Q_2$

保持冰箱在 5°C 至 20°C 之间运转，
每天需做功

$$W = \frac{Q_2}{e} = 0.2 \times 10^7 \text{ W}$$

功率 $P = \frac{W}{t} = \frac{0.2 \times 10^7}{24 \times 3600} \text{ W} = 23 \text{ W}$

2、 摩尔数为 $\nu=1$ 的单原子分子理想气体，
从初态A出发，经历如图循环过程，求：

(0) 分别确定三个端点的状态参量(P, V, T, E)。

分别在P-T图和V-T图上化出循环过程。

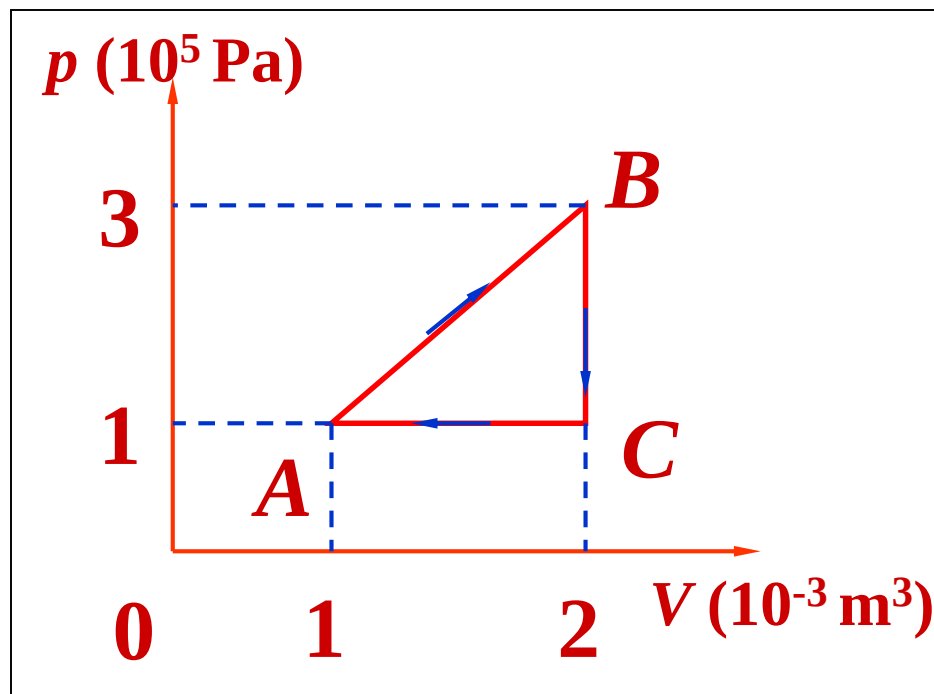
(1) 各过程系统做功

W、内能变化 ΔE 、吸
热量**Q**和摩尔热容。

(2) 整个循环过程

系统对外作的总功
及净吸热。

(3) 该循环的效率。



解 **A—B**

$$W_{AB} = \frac{1}{2}(p_A + p_B)(V_B - V_A) = 200 \text{ J}$$

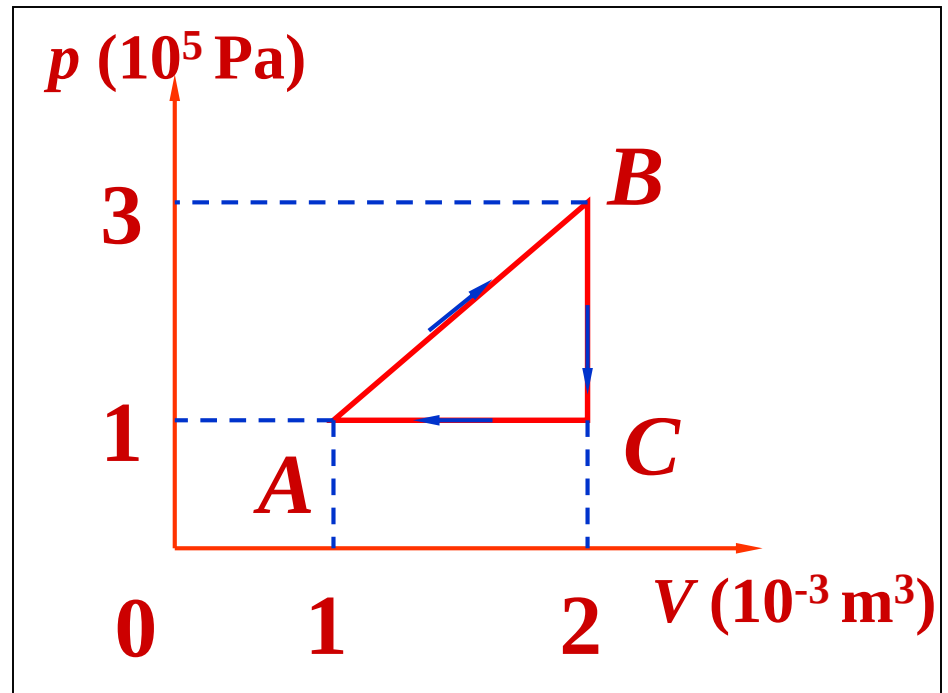
$$\Delta E_{AB} = \nu C_{V,m}(T_B - T_A)$$

$$= \nu \frac{3}{2} R(T_B - T_A)$$

$$= \frac{3}{2}(p_B V_B - p_A V_A) = 750 \text{ J}$$

$$Q_{AB} = \Delta E_{AB} + W_{AB} = 950 \text{ J}$$

$$Q_{AB} = \nu C_m \Delta T$$



$B—C$ 等容

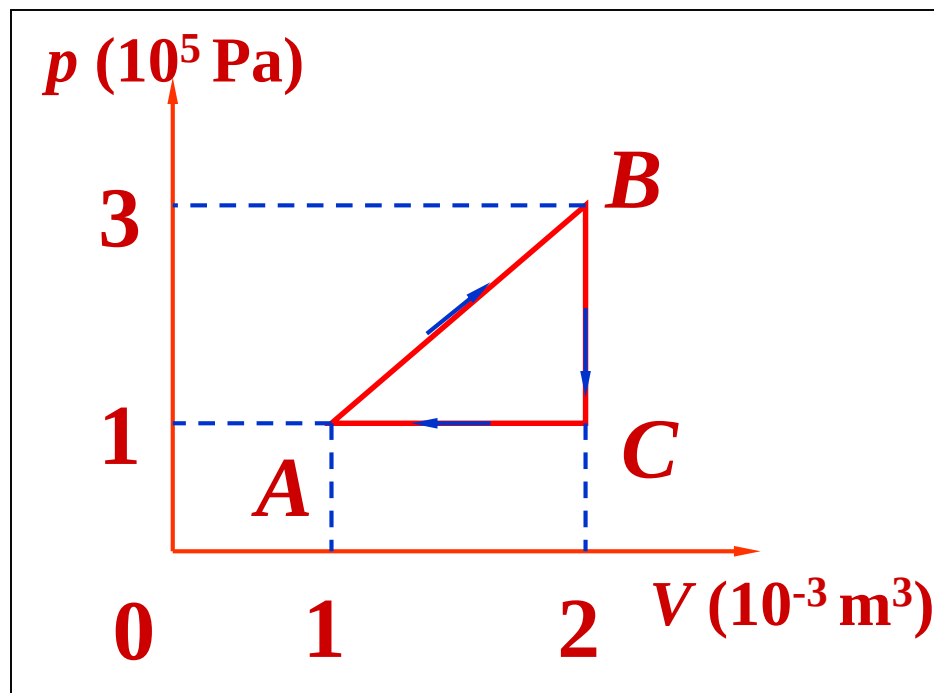
$$W_{BC} = 0$$

$$\Delta E_{BC} = \frac{3}{2}(p_C V_C - p_B V_B)$$

$$= -600 \text{ J}$$

$$Q_{BC} = \Delta E_{BC} + W_{BC}$$

$$= -600 \text{ J}$$



C—A 等压

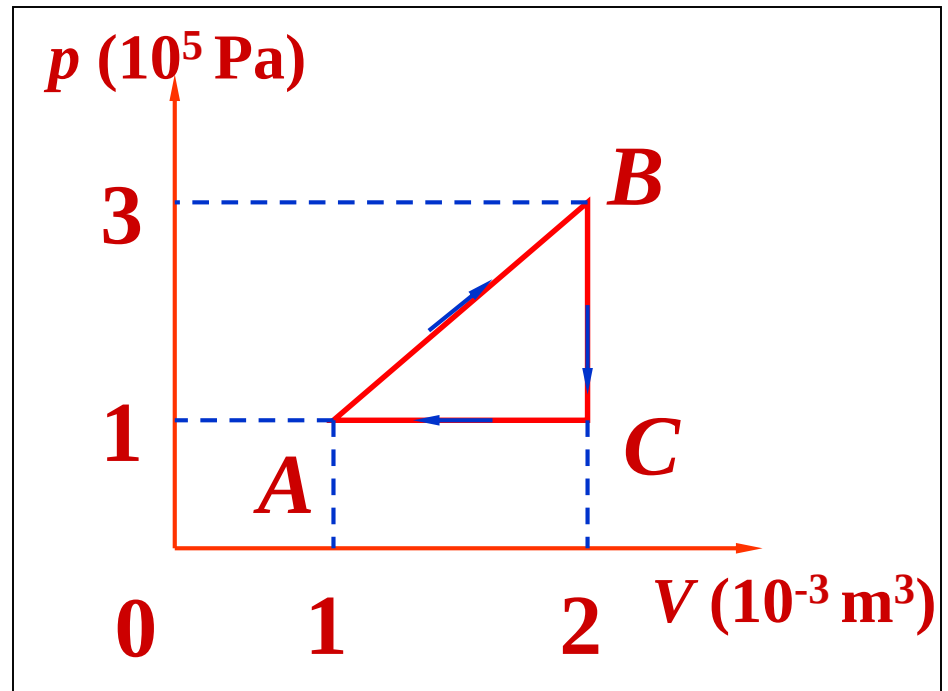
$$W_{CA} = p_A(V_A - V_C) = -100 \text{ J}$$

$$\Delta E_{CA} = \frac{3}{2}(p_A V_A - p_C V_C)$$

$$= -150 \text{ J}$$

$$Q_{CA} = \Delta E_{CA} + W_{CA}$$

$$= -250 \text{ J}$$



$$\eta = \frac{W}{Q_{\text{吸}}} = \frac{W}{Q_{AB}}$$

$$= \frac{100}{950} = 10.5\%$$

